Homework 2

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Exercise 1. Worked example for an Elliptic PDE
Consider the Helmholtz equation:

$$u_{xx} + u_{yy} + u = 0 \quad \text{for } x \in (0,3), y \in (0,4),$$

with Dirichlet boundary conditions:

$$u(x=0,y) = 0, \quad u(x=3,y) = 30y, \quad u(x,y=0) = 0, \quad u(x,y=4) = 40x.$$

Consider the following discretisation based on the five-point Laplacian:

$$\frac{U_{m-1,p} + U_{m+1,p} - 4U_{m,p} + U_{m,p-1} + U_{m,p+1}}{(\Delta x)^2} + U_{m,p} = 0 \quad \text{for } m \in \{1,\dots,M-1\}, p \in \{1,\dots,P-1\},$$

with the boundary conditions enforced via

$$\begin{aligned} U_{0,p} = 0, \quad U_{M,p} = 30p\Delta x \qquad & \text{for } p \in \{1,\dots,P-1\}, \\ U_{m,0} = 0, \quad U_{m,P} = 40m\Delta x \qquad & \text{for } m \in \{1,\dots,M-1\}. \end{aligned}$$

Here $U_{m,p}$ is the discrete solution at $(x,y) = (m\Delta x, p\Delta x)$, where $\Delta x = \frac{3}{M} = \frac{4}{P}$ with M and P positive integers.

Now, take the case $M = 3$, $P = 4$ (i.e. $\Delta x = 1$). Show that upon discretisation one obtains the linear system:

$$A \begin{bmatrix} U_{1,1} \\ U_{2,1} \\ U_{1,2} \\ U_{2,2} \\ U_{1,3} \\ U_{2,3} \end{bmatrix} = \begin{bmatrix} 0 \\ -30 \\ 0 \\ -60 \\ -40 \\ -170 \end{bmatrix},$$

where A is a 6×6 matrix which should be found.

Solution. The exercise told us to discretize the Helmholtz equation and resulting linear system $A\underline{u} = \underline{b}$. Using the given boundary conditions, we can write the grid parameters as follows:

$$M = 3, P = 4 \implies \Delta x = 3/3 = 1, \Delta y = 4/4 = 1.$$

Now since we are given the discretization formula, we can substitute $\Delta x = 1$ into it, and rearrange it:

$$\frac{U_{m-1,p} + U_{m+1,p} - 4U_{m,p} + U_{m,p-1} + U_{m,p+1}}{(\Delta x)^2} + U_{m,p} = 0 \implies$$

$$\implies -3U_{m,p} + U_{m-1,p} + U_{m+1,p} + U_{m,p-1} + U_{m,p+1} = 0.$$

For internal nodes (m,p), we have $m \in \{1,2\}$ and $p \in \{1,2,3\}$, giving us a total of 6 internal nodes.
Next sensitive step is the application of boundary conditions. We have:

$$U_{0,p} = 0, \quad U_{m,0} = 0 \tag{1}$$

$$u = 30y \text{ at } M = 3 \implies \begin{cases} U_{3,1} = 30 \cdot 1 = 30 \\ U_{3,2} = 30 \cdot 2 = 60 \\ U_{3,3} = 30 \cdot 3 = 90 \end{cases} \qquad u = 40x \text{ at } P = 4 \implies \begin{cases} U_{1,4} = 40 \cdot 1 = 40 \\ U_{2,4} = 40 \cdot 2 = 80 \\ U_{3,4} = 40 \cdot 3 = 120 \end{cases} \tag{2}$$

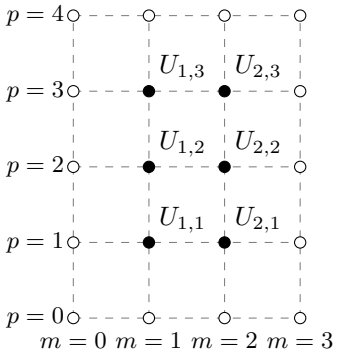
Finally, we can write the final equations for each internal node:

$$\begin{aligned} \text{For}(m,p) = (1,1) \qquad & -3U_{1,1} + U_{2,1} + U_{1,2} = 0 \\ \text{For}(m,p) = (2,1) \qquad & U_{1,1} - 3U_{2,1} + U_{2,2} = -30 \\ \text{For}(m,p) = (1,2) \qquad & U_{1,1} - 3U_{1,2} + U_{2,2} + U_{1,3} = 0 \\ \text{For}(m,p) = (2,2) \qquad & U_{2,1} + U_{1,2} - 3U_{2,2} + U_{2,3} = -60 \\ \text{For}(m,p) = (1,3) \qquad & U_{1,2} - 3U_{1,3} + U_{2,3} = -40 \\ \text{For}(m,p) = (2,3) \qquad & U_{2,2} + U_{1,3} - 3U_{2,3} = -170 \end{aligned} \tag{3}$$

Now we can write the system in matrix form $A\underline{u} = \underline{b}$ as requested, where:

$$A = \begin{bmatrix} \mathbf{-3} & 1 & 1 & 0 & 0 & 0 \\ 1 & \mathbf{-3} & 0 & 1 & 0 & 0 \\ 1 & 0 & \mathbf{-3} & 1 & 1 & 0 \\ 0 & 1 & 1 & \mathbf{-3} & 0 & 1 \\ 0 & 0 & 1 & 0 & \mathbf{-3} & 1 \\ 0 & 0 & 0 & 1 & 1 & \mathbf{-3} \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} U_{1,1} \\ U_{2,1} \\ U_{1,2} \\ U_{2,2} \\ U_{1,3} \\ U_{2,3} \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} 0 \\ -30 \\ 0 \\ -60 \\ -40 \\ -170 \end{bmatrix}. \tag{4}$$

We have thus found the required matrix $A = 6 \times 6$. For clarity, we also provide a sketch of the grid defining the internal and boundary nodes:



□

Exercise 2. Worked example for Crank-Nicolson method applied to the heat equation
Consider the heat equation:

$$u_t = u_{xx} \quad \text{for } x \in (0,1), t \in (0,T),$$

with boundary conditions given by

$$u(x = 0, t) = u(x = 1, t) = 0,$$

and the initial condition

$$u(x, t = 0) = x(1 - x).$$

Suppose the problem is solved using the Crank–Nicolson method, with $\Delta t = 0.02$ and $\Delta x = 0.2$. There are therefore 6 nodes in the spatial variable x: $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Show that

$$A \begin{bmatrix} U_1^1 \\ U_2^1 \\ U_3^1 \\ U_4^1 \end{bmatrix} = B \begin{bmatrix} U_1^0 \\ U_2^0 \\ U_3^0 \\ U_4^0 \end{bmatrix} = \underline{b},$$

with U_m^n denoting the fully discrete approximation of $u(x = m\Delta x, t = n\Delta t)$. Here, A and B are 4×4 matrices which should be found. Please also calculate the vector $\underline{b}$.

Solution. We start using method of lines to discretize the heat equation using Crank-Nicolson method.

1. Discretization of the space $x \in (0, 1)$ with step $\Delta x = 0.2$. This gives us 6 nodes: $m \in \{0, 1, 2, 3, 4, 5\}$. However, since we have Dirichlet boundary conditions at $x = 0$ and $x = 1$, we will only consider the internal nodes:

$$x_m = m\Delta x, \quad m \in \{1, 2, 3, 4\}, \quad \Delta x = 0.2.$$

2. Replace u_{xx} with the central difference quotient, resulting in the semi-discrete equation:

$$\frac{dU_m}{dt} = \frac{U_{m-1} - 2U_m + U_{m+1}}{(\Delta x)^2}$$

3. Discretization of the time using Crank-Nicolson method (trapezoidal rule):

$$\frac{U_m^{n+1} - U_m^n}{\Delta t} = \frac{1}{2}\left[\frac{U_{m-1}^{n+1} - 2U_m^{n+1} + U_{m+1}^{n+1}}{(\Delta x)^2} + \frac{U_{m-1}^n - 2U_m^n + U_{m+1}^n}{(\Delta x)^2}\right]$$

4. Rearranging the above equation using $r = \frac{\Delta t}{(\Delta x)^2}$ for clarity:

$$-r/2\, U_{m-1}^{n+1} + (1 + r)U_m^{n+1} - r/2\, U_{m+1}^{n+1} = r/2\, U_{m-1}^n + (1 - r)U_m^n + r/2\, U_{m+1}^n$$

5. Substituting in r the given values $\Delta t = 0.02$ and $\Delta x = 0.2$:

$$r = \frac{\Delta t}{(\Delta x)^2} = \frac{0.02}{(0.2)^2} = 0.5$$

$$-0.25U_{m-1}^{n+1} + 1.5U_m^{n+1} - 0.25U_{m+1}^{n+1} = 0.25U_{m-1}^n + 0.5U_m^n + 0.25U_{m+1}^n$$

6. Now thanks to the boundary conditions $U_0^n = U_0^{n+1} = 0$ for all n and $U_5^n = U_5^{n+1} = 0$ for all n, we can write the equations for $m \in \{1, 2, 3, 4\}$:

$$\begin{aligned}
&\text{For } m = 1: \quad -0.25 \cdot (0) + 1.5U_1^{n+1} - 0.25U_2^{n+1} = 0.25 \cdot (0) + 0.5U_1^n + 0.25U_2^n \\
&\text{For } m = 2: \quad -0.25U_1^{n+1} + 1.5U_2^{n+1} - 0.25U_3^{n+1} = 0.25U_1^n + 0.5U_2^n + 0.25U_3^n \\
&\text{For } m = 3: \quad -0.25U_2^{n+1} + 1.5U_3^{n+1} - 0.25U_4^{n+1} = 0.25U_2^n + 0.5U_3^n + 0.25U_4^n \\
&\text{For } m = 4: \quad -0.25U_3^{n+1} + 1.5U_4^{n+1} - 0.25 \cdot (0) = 0.25U_3^n + 0.5U_4^n + 0.25 \cdot (0)
\end{aligned} \tag{5}$$

We can now write the system in matrix form $A\underline{U}^{n+1} = B\underline{U}^n = \underline{b}$ as requested, where:

$$\begin{bmatrix} 1.5 & -0.25 & 0 & 0 \\ -0.25 & 1.5 & -0.25 & 0 \\ 0 & -0.25 & 1.5 & -0.25 \\ 0 & 0 & -0.25 & 1.5 \end{bmatrix} \begin{bmatrix} U_1^{n+1} \\ U_2^{n+1} \\ U_3^{n+1} \\ U_4^{n+1} \end{bmatrix} = \begin{bmatrix} 0.5 & 0.25 & 0 & 0 \\ 0.25 & 0.5 & 0.25 & 0 \\ 0 & 0.25 & 0.5 & 0.25 \\ 0 & 0 & 0.25 & 0.5 \end{bmatrix} \begin{bmatrix} U_1^n \\ U_2^n \\ U_3^n \\ U_4^n \end{bmatrix} = \underline{b} \tag{6}$$

Finally, we need to calculate the vector $\underline{b}$ at time step $n = 0$ using $\underline{b} = B \cdot U^0$ where U^0 is the initial condition $u(x, t = 0) = x(1 - x)$ and B is the matrix found above. We have:

$$U^0 = \begin{bmatrix} 0.16 & 0.24 & 0.36 & 0.48 \end{bmatrix} \implies \underline{b} = \begin{bmatrix} 0.14 & 0.22 & 0.22 & 0.14 \end{bmatrix}$$

□

Exercise 3. Stability of Crank-Nicolson method for the heat equation
Consider the heat equation:

$$u_t = \kappa u_{xx} \quad \text{for } x \in (0,1), t \in (0,T),$$

coupled with an appropriate initial condition and periodic boundary conditions, where $\kappa > 0$ and $T > 0$. Consider the Crank–Nicolson method

$$\frac{U_m^{n+1} - U_m^n}{\Delta t} = \frac{1}{2}\left[\kappa \frac{U_{m-1}^n - 2U_m^n + U_{m+1}^n}{(\Delta x)^2} + \kappa \frac{U_{m-1}^{n+1} - 2U_m^{n+1} + U_{m+1}^{n+1}}{(\Delta x)^2}\right]$$

$$\text{for } m \in \{1, \ldots, M-1\}, n \in \{0, \ldots, N-1\},$$

Here U_m^n is the discrete solution at $x = m\Delta x$ and $t = n\Delta t$, where $\Delta x = \frac{1}{M}$ and $\Delta t = \frac{T}{N}$ with M and N positive integers. [You may assume that the initial condition and boundary conditions are appropriately discretised.]
Apply von Neumann analysis, and show that the discretisation is *unconditionally stable*, that is the method is stable for any positive Δx and Δt.

Solution. First, we assume the solution can be expressed as a Fourier mode: $U_m^n = \xi^n e^{ikm\Delta x}$

$$\begin{aligned}\frac{\xi^{n+1}e^{ikm\Delta x} - \xi^n e^{ikm\Delta x}}{\Delta t} =& \frac{1}{2}\Bigg[\kappa \frac{\xi^n e^{ik(m-1)\Delta x} - 2\xi^n e^{ikm\Delta x} + \xi^n e^{ik(m+1)\Delta x}}{(\Delta x)^2} \\ &+ \kappa \frac{\xi^{n+1} e^{ik(m-1)\Delta x} - 2\xi^{n+1} e^{ikm\Delta x} + \xi^{n+1} e^{ik(m+1)\Delta x}}{(\Delta x)^2}\Bigg]\end{aligned} \tag{7}$$

Dividing both sides by $\xi^n e^{ikm\Delta x}$, multiply by Δt and rearranging, we get:

$$\xi - 1 = \frac{\kappa \Delta t}{2(\Delta x)^2}\left[\left(e^{-ik\Delta x} - 2 + e^{ik\Delta x}\right) + \xi\left(e^{-ik\Delta x} - 2 + e^{ik\Delta x}\right)\right] \tag{8}$$

We can simplify the terms in parentheses using Euler's formula: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ and by trigonometric identity: $1 - \cos(\theta) = 2\sin^2\left(\frac{\theta}{2}\right)$

$$e^{-ik\Delta x} - 2 + e^{ik\Delta x} = 2\cos(k\Delta x) - 2 = -2(1 - \cos(k\Delta x)) = -4\sin^2\left(\frac{k\Delta x}{2}\right)$$

By that we have:

$$\xi - 1 = -2\frac{\kappa\Delta t}{(\Delta x)^2}\left[\sin^2\left(\frac{k\Delta x}{2}\right) + \xi\sin^2\left(\frac{k\Delta x}{2}\right)\right] \implies \xi - 1 = -2\frac{\kappa\Delta t}{(\Delta x)^2}\sin^2\left(\frac{k\Delta x}{2}\right)(1+\xi) \tag{9}$$

Rearranging for ξ:

$$\xi = \frac{1 - 2\frac{\kappa\Delta t}{(\Delta x)^2}\sin^2\left(\frac{k\Delta x}{2}\right)}{1 + 2\frac{\kappa\Delta t}{(\Delta x)^2}\sin^2\left(\frac{k\Delta x}{2}\right)} \tag{10}$$

Now we need to check $|\xi| \leq 1$ for stability:

$$|\xi| = \left|\frac{1 - 2\frac{\kappa\Delta t}{(\Delta x)^2}\sin^2\left(\frac{k\Delta x}{2}\right)}{1 + 2\frac{\kappa\Delta t}{(\Delta x)^2}\sin^2\left(\frac{k\Delta x}{2}\right)}\right| \leq 1 \tag{11}$$

But since $\kappa > 0$, $\Delta t > 0$, $(\Delta x)^2 > 0$ and $\sin^2\left(\frac{k\Delta x}{2}\right) \geq 0$, the denominator is always positive and greater than the **modulus** of the numerator. Therefore, the inequality holds for any positive Δx and Δt, proving that the Crank-Nicolson method is unconditionally stable as requested. □

Exercise 4. A scheme for the advection equation
Consider the advection equation:

$$u_t + vu_x = 0 \quad \text{for } x \in (0,1), t \in (0,T),$$

where v is a constant and $T > 0$, subject to an initial condition

$$u(x, t = 0) = P(x),$$

and with periodic boundary conditions in the x-dimension.

This can be discretised to yield

$$\frac{U_m^{n+1} - U_m^n}{\Delta t} + v\frac{U_m^{n+1} - U_{m-1}^{n+1}}{\Delta x} = 0 \quad \text{for } m \in \{0, \ldots, M-1\}, n \in \{0, \ldots, N-1\},$$

where U_m^n is the discrete solution at $x = m\Delta x$ and $t = n\Delta t$, where $\Delta x = \frac{1}{M}$ and $\Delta t = \frac{T}{N}$ with M and N positive integers. The initial condition is applied via

$$U_m^0 = P(x = m\Delta x) \quad \text{for } m \in \{0, \ldots, M-1\},$$

and periodic boundary conditions via

$$U_{-1}^n = U_{M-1}^n \quad \text{for } n \in \{1, \ldots, N\}.$$

4.1 Show that, if the Courant number

$$C = \frac{v\Delta t}{\Delta x},$$

is held constant, then the discretisation is consistent (in an appropriate norm). You may assume here that $v \neq 0$.
[*Hint: Examine the local truncation error, fixing Δx in terms of Δt. Then adapt the working of Lecture 6 at each time point to prove consistency.*]

4.2 Apply von Neumann analysis, and find a sufficient condition, expressed in terms of the Courant number C, for stability (in the 2-norm).

Solution. 4.1 Solution: As suggested in the hint, we start by examining the local truncation error.

From the discretisation:

$$\frac{U_m^{n+1} - U_m^n}{\Delta t} + v\frac{U_m^{n+1} - U_{m-1}^{n+1}}{\Delta x} = 0 \quad \text{for } m \in \{0, \ldots, M-1\}, n \in \{0, \ldots, N-1\} \tag{12}$$

We can express the local truncation error τ_m^n substituting exact solution $u(x_m, t_n)$ into the discretisation:

$$\tau_m^n = \frac{u(x_m, t_{n+1}) - u(x_m, t_n)}{\Delta t} + v\frac{u(x_m, t_{n+1}) - u(x_{m-1}, t_{n+1})}{\Delta x}$$

It is convenient to expand terms using Taylor series expansions around the point (x_m, t_{n+1}). Firstly with respect to time t:

$$u(x_m, t_n) = u(x_m, t_{n+1} - \Delta t) = u(x_m, t_{n+1}) - \Delta t u_t(x_m, t_{n+1}) + \frac{(\Delta t)^2}{2}u_{tt}(x_m, t_{n+1}) - \cdots$$

$$\implies \frac{u(x_m, t_{n+1}) - u(x_m, t_n)}{\Delta t} = u_t(x_m, t_{n+1}) - \frac{\Delta t}{2}u_{tt}(x_m, t_{n+1}) + O(\Delta t^2)$$

Next, we expand with respect to space x:

$$u(x_{m-1}, t_{n+1}) = u(x_m - \Delta x, t_{n+1}) = u(x_m, t_{n+1}) - \Delta x u_x(x_m, t_{n+1}) + \frac{(\Delta x)^2}{2} u_{xx}(x_m, t_{n+1}) - \cdots$$

$$\implies \frac{u(x_m, t_{n+1}) - u(x_{m-1}, t_{n+1})}{\Delta x} = u_x(x_m, t_{n+1}) - \frac{\Delta x}{2} u_{xx}(x_m, t_{n+1}) - O(\Delta x^2)$$

Now substituting these expansions back into the expression for τ_m^n and rearranging:

$$\tau_m^n = \left(u_t(x_m, t_{n+1}) - \frac{\Delta t}{2} u_{tt}(x_m, t_{n+1}) + O(\Delta t^2) \right) + v \left(u_x(x_m, t_{n+1}) - \frac{\Delta x}{2} u_{xx}(x_m, t_{n+1}) + O(\Delta x^2) \right)$$

$$= u_t(x_m, t_{n+1}) + v u_x(x_m, t_{n+1}) - \frac{\Delta t}{2} u_{tt}(x_m, t_{n+1}) - \frac{v \Delta x}{2} u_{xx}(x_m, t_{n+1}) + O(\Delta t^2) + O(\Delta x^2)$$

From the advection equation, we know that $u_t + v u_x = 0$, so the first two terms cancel out, leaving us with:

$$\tau_m^n = -\frac{\Delta t}{2} u_{tt}(x_m, t_{n+1}) - \frac{v \Delta x}{2} u_{xx}(x_m, t_{n+1}) + O(\Delta t^2) + O(\Delta x^2)$$

At this point, from Courant number definition like a costant $C = \frac{v \Delta t}{\Delta x}$, we can express Δx in terms of Δt:

$$\Delta x = \frac{v \Delta t}{C} \implies \tau_m^n = -\frac{\Delta t}{2} u_{tt}(x_m, t_{n+1}) - \frac{v^2 \Delta t}{2C} u_{xx}(x_m, t_{n+1}) + O(\Delta t^2)$$

Finally, we arrive at the expression for the local truncation error only in terms of Δt:

$$\tau_m^n = -\Delta t \left(\frac{1}{2} u_{tt}(x_m, t_{n+1}) + \frac{v^2}{2C} u_{xx}(x_m, t_{n+1}) \right)$$

That allows us to analize the consistency of the scheme. As $\Delta t \to 0$, the limit of the local truncation error $\lim_{\Delta t \to 0} |\tau_m^n| \to 0$ for all m and n. Therefore, the discretisation is consistent in the appropriate norm, as required.

4.2 Solution: First of all we substitute the courant number into the discretisation scheme given above, this can be done by multiplying the entire equation by Δt:

$$\frac{U_m^{n+1} - U_m^n}{\Delta t} + v \frac{U_m^{n+1} - U_{m-1}^{n+1}}{\Delta x} = 0 \implies U_m^{n+1} - U_m^n + C(U_m^{n+1} - U_{m-1}^{n+1}) = 0$$

Arranging for U_m^{n+1}:

$$U_m^{n+1}(1 + C) - C U_{m-1}^{n+1} = U_m^n$$

Now we can apply von Neumann analysis by assuming the solution can be expressed as a Fourier mode:

- $U_m^{n+1} \to \xi^{n+1} e^{ikm\Delta x}$
- $U_{m-1}^{n+1} \to \xi^{n+1} e^{ik(m-1)\Delta x} = \xi^{n+1} e^{ikm\Delta x} e^{-ik\Delta x}$
- $U_m^n \to \xi^n e^{ikm\Delta x}$

Substituting these into the rearranged discretisation:

$$\xi^{n+1} e^{ikm\Delta x}(1 + C) - C \xi^{n+1} e^{ikm\Delta x} e^{-ik\Delta x} = \xi^n e^{ikm\Delta x}$$

At this point we can divide both sides by $\xi^n e^{ikm\Delta x}$ and evaluate for ξ using euler formula $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ considering that $\theta = k\Delta x$:

$$\xi(1 + C - Ce^{-ik\Delta x}) = 1 \implies \xi = \frac{1}{1 + C - Ce^{-ik\Delta x}} \implies \xi = \frac{1}{1 + C - C(\cos(\theta) + i\sin(\theta))}$$

To check for stability in norm 2, we need to ensure that $|\xi| \leq 1$, so the denominator must be greater than or equal to 1. We can calculate the modulus of the denominator:

$$\underbrace{1 + C - C\cos(\theta)}_{\text{real part}} + i \underbrace{C\sin(\theta)}_{\text{imaginary part}} \geq 1 \implies |...|^2 \geq 1$$

Calculating the modulus with some algebraic manipulations:

$$\begin{aligned}(1 + C - C\cos(\theta))^2 + (C\sin(\theta))^2 = ... &= 1 + 2C + 2C^2 - 2C^2\cos(\theta) \\ &= 1 + 2C(1 + C)(1 - \cos(\theta)) \geq 1 \\ &\implies 2C(1 + C)(1 - \cos(\theta)) \geq 0\end{aligned}$$

In that form we can see that for all θ, $1 - \cos(\theta) \geq 0$. Therefore, the inequality holds if $C(1 + C) \geq 0$. This is true when $C \geq 0$ or $C \leq -1$. We can conclude that a sufficient condition for stability in norm 2 is:

$$C \geq 0 \quad \text{or} \quad C \leq -1$$

In a physical context, in particular in this exercise, we know from text that U_{m-1} implies $v > 0$, so the condition $C \geq 0$ is the solution for this advection equation. □